

Paired Prototype Descriptive Document : Leap of Faith

Team Members

Name	Email	GitHub Username
Samarth Narahari Mandagere	mandager@usc.edu	MNSamarth
Xiufang Chen	xiufangc@usc.edu	repoRookie

Logline

(Vertical Platformer + Timed World Flips)

Leap Of Faith is a vertical platformer where warned, game-controlled 180-degree world rotations turn climbs into dangerous descents, forcing a constantly hopping player to anticipate each flip and prepare safe landings on the way to the exit.

Genre Research

We studied three popular **platforming and gravity-manipulation** titles:

1. *Doodle Jumps*
2. *And Yet It Moves*
3. *VVVVVV*

Tropes:

When examining vertical platformers and gravity-manipulation games, we find several recurring design elements. These games use simple movement controls while creating difficulty through precise positioning, environmental hazards, and altered gravity. Vertical progression encourages players to move between platforms, while automatic jumping or gravity-based movement limits direct control and makes timing essential. Changing the world's orientation transforms familiar spaces into new traversal challenges, requiring players to anticipate where they will land. As levels progress, tighter platforms, more hazards, and increasingly complex layouts demand greater precision and adaptation.

- **Simple Movement Controls:** Players use a small set of controls, allowing the challenge to come from timing and positioning.
- **Vertical Progression:** Levels encourage upward or downward movement through platforms and obstacles.
- **Collectibles and Optional Challenges:** Stars reward exploration and encourage players to take riskier routes beyond simply reaching the exit.
- **Gravity Manipulation:** Gravity changes direction or the environment rotates, altering how players navigate the level.
- **Precision Platforming:** Players must carefully control their movement to land safely on small or distant platforms.

- **Environmental Hazards:** Spikes, gaps, and other dangers punish to loses the game.
- **Timing and Anticipation:** Players must predict jumps, gravity changes, and safe landing positions.
- **Repeated Attempts:** Quick failure and restarting encourage players to learn the level through practice.
- **Difficulty Scaling:** Later sections introduce narrower platforms, more hazards, and more complicated routes.

Twist:

Our core twist is **Timed World Rotation & Automatic Hopping:**

- **Automatic Hopping:** The character jumps continuously, so the player must steer in midair and carefully plan each landing.
- **Timed World Rotation:** After a warning, the game automatically rotates the entire level 180 degrees, including its platforms, hazards, and exit.
- **Jump Boost:** Players can trigger a higher jump to reach distant platforms or reposition themselves before the world flips.

How the Twist Innovates on Leap of Faith

1. A Complete, Game-Controlled World Flip:

- In And Yet It Moves, the player rotates the world in 90-degree turns.
- In our game, the world automatically flips completely upside down with a 180-degree rotation after a warning.

2. Climbing Becomes Falling:

- An upward route instantly becomes a dangerous descent after each flip.
- Platforms must serve as both launch points and potential landing zones.
- Some stars are placed in areas that become reachable only after the world flips, making rotation part of exploration rather than only a hazard.

3. Warning-Based Preparation:

- A warning gives the player a brief opportunity to react before the world flips.
- Players must choose between continuing toward the exit or moving into a safer position for the upcoming descent.

Together, these mechanics combine **precision platforming**, **anticipation**, and **route adaptation**, turning the same level into two opposite traversal challenges.

Prototype Description

This prototype is a Unity 2D vertical platformer featuring player-controlled movement, jumping, collectible stars, and timed 180-degree world rotations. After a warning, the entire level—including its platforms, hazards, collectibles, and exit—flips upside down,

turning upward climbs into dangerous descents. Each level contains five stars, with some placed in areas that can only be reached by using the world flip. Players must anticipate each rotation, land safely, collect as many stars as possible, and reach the exit.

Mechanic Matrices

Twist & Mechanics Matrix

Mechanic	Description	Interaction with Twist	Affected Genre Elements	Type of Genre Innovation
Timed 180° World Rotation	After a set interval, the entire level flips upside down, changing the orientation of platforms, hazards, collectibles, and the exit.	Turns upward progression into a downward traversal challenge and forces players to reinterpret the same level layout.	Gravity Manipulation; Platform Traversal; Level Progression	Core Mechanic Innovation
Flip Warning	A visual warning appears shortly before each world rotation.	Gives players a brief opportunity to find a safe position and prepare their next landing.	Timing; Player Feedback; Precision Platforming	Strategic Twist
Flip-Dependent Stars	Each level contains five collectible stars, including stars placed in pockets that require a world flip to reach.	Makes the rotation a tool for exploration and rewards players for understanding both level orientations.	Collectibles; Exploration; Optional Challenges	Level-Design Innovation
Orientation-Sensitive Hazards	Some platforms have dangerous red undersides that become exposed or repositioned after a flip.	Changes which surfaces and routes are safe in each orientation, requiring players to adapt after every rotation.	Environmental Hazards; Route Planning; Difficulty Scaling	Environmental Twist

GitHub Repository

[Leap of Faith GitHub Repository](#)

WebGL Playable Link

[Unity Web Player | Leap of Faith](#)

Gameplay Video

[Leap of Faith Gameplay Video](#)

Individual Contributions

Samarth Narahari Mandagere	<i>Co-developed the game concept and gameplay structure; implemented the prototype's main gameplay systems, including player movement, jumping, timed world rotation, and supporting mechanics; built and deployed the playable game.</i>
Xiufang Chen	<i>Co-developed the game concept and gameplay structure; designed two gameplay levels; conducted comparative genre research on; prepared the submission documentation and discord marketing description; and recorded the gameplay demonstration video.</i>

Diagram/Sketch

